

Fuzzy-Based Crime Prevention Strategies for Urban Safety

Mid-term presentation for the Seminar on Urban Computing

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Abstract

Our goal is to introduce a fuzzy-based crime prevention platform designed to enhance urban safety by adapting to real-time data. Unlike existing methods, this system leverages fuzzy logic to dynamically prevent crimes and public disturbances(a person breaking things, being too loud, etc.), by processing various inputs, including hotspot reports from any media(for example ongoing protests), as well as feedback from both governmental bodies and citizens. By encouraging community involvement and promoting fairness in resource allocation, the platform offers a human-centered approach designed to improve urban safety.





Introduction to the problem

- We all know it is hard to prevent crimes from happening, and even harder to predict them.

However, we can always use our human sense to improve the security and safety in our living environment.

- Additionally, Switzerland is one of the safest countries in the world, and the crime rate is very low compared to the rest of the world.

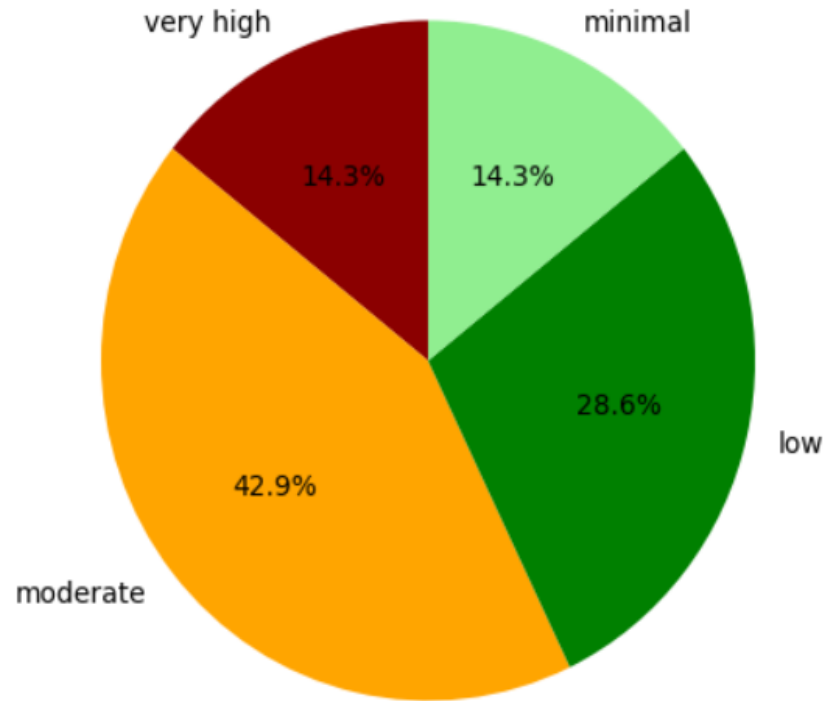
Nevertheless, our goal is to strengthen the urban safety by making the authorities and official bodies have an easier approach to their job.

Elaborating on the pains

- Growing crime rates and unpredictable patterns in urban areas.
- Current prevention methods are rigid, reactive, and lack adaptability.
- Human-centered solutions are missing in today's systems.
- Key issues: surveillance overload, biased algorithms, and inefficient resource distribution.



Crime Risk Distribution for Bern



How Fuzzy Logic Works in Urban Safety

- Fuzzy logic enables nuanced decision-making by classifying crime risk on a spectrum, rather than as simply “safe” or “dangerous”.
- Our platform evaluates crime severity, frequency, and other factors to predict risks more accurately. This leads to better interventions and improved urban safety.
- For example, a neighborhood might be classified as “relatively safe” or “moderately risky” based on real-time data, offering a more flexible approach to crime prevention.

The Pitch Deck (Eureka!)

Our Approach

- Our platform will use fuzzy-based algorithms to process ambiguous data, allowing for **flexible crime prediction** and **dynamic resource management**.
- Engages the community, providing both law enforcement and citizens a voice in shaping safety strategies.
- Emphasizes social equity by avoiding biased predictions and over-policing.

Artifact Features

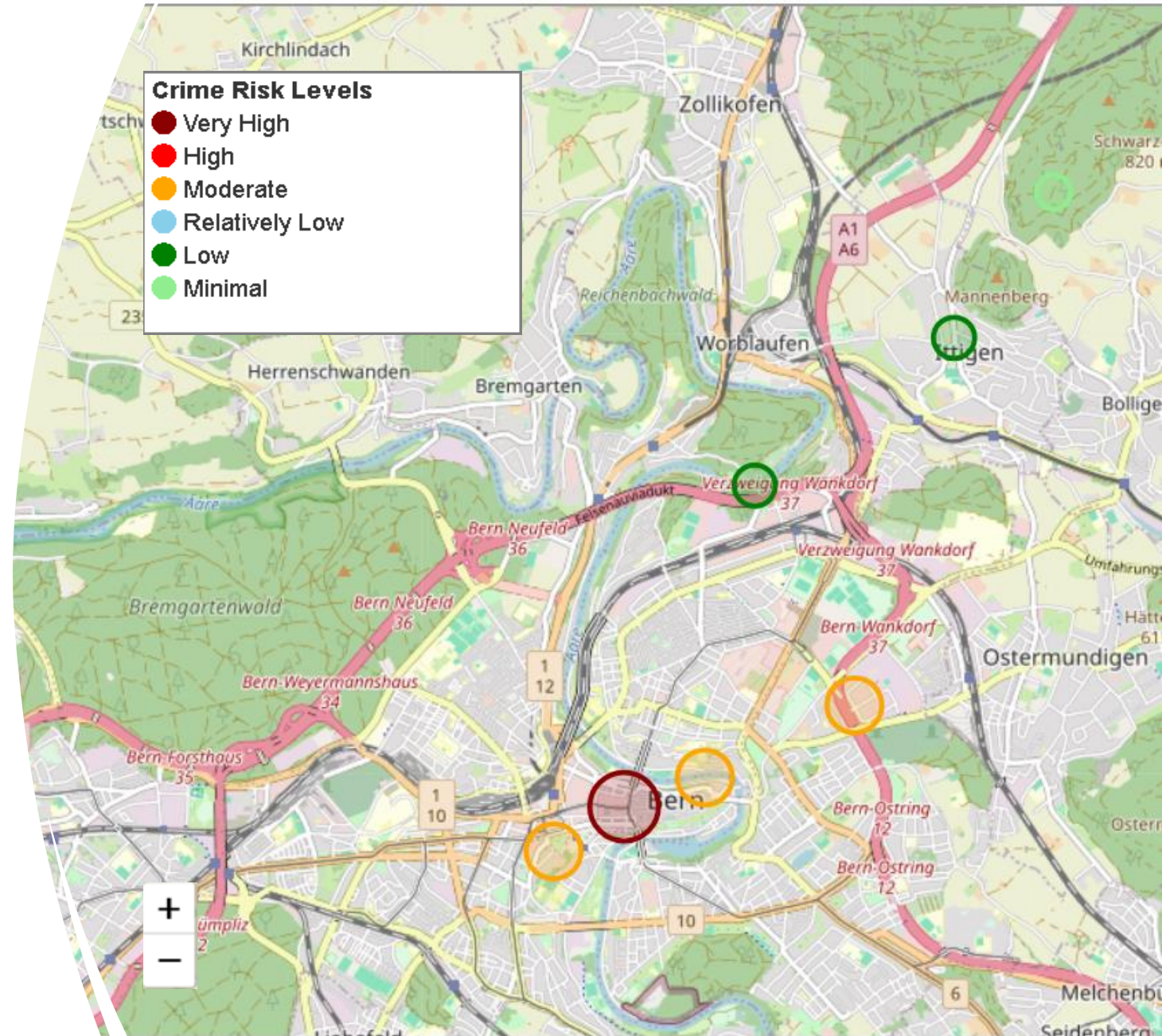
- **Risk Assessment:** Fuzzy logic can analyze various risk factors (e.g., previous crime rates, time & seasonality and event schedules) to identify areas at higher risk for crime. This allows for targeted interventions.
- **Predictive Policing:** By utilizing fuzzy algorithms, law enforcement can better predict potential crime hotspots based on incomplete or uncertain data, leading to more effective resource allocation.
- **Community Engagement:** Fuzzy logic can help process qualitative feedback from community surveys, allowing police to understand public concerns more effectively and adjust strategies accordingly.
- **Resource Optimization:** Fuzzy logic can help optimize patrol routes and schedules based on various parameters like time of day, local events, and historical data, ensuring a more efficient use of police resources.

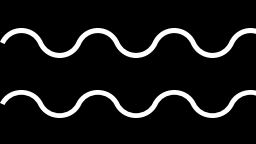
Artifact Goal: Create an easy-to-use platform that law enforcement, urban planners, and communities can utilize to make better decisions in crime prevention.

Crime Hotspot Map for Bern

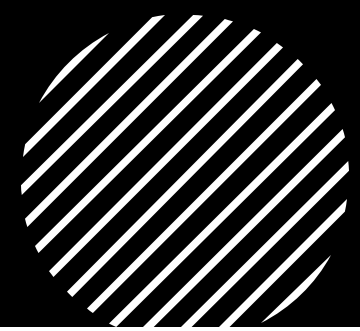
Story

Quote: “In cities of the future, safety isn’t about force—it’s about flexibility.”





Takeaway & Next Steps



Design a risk assessment system that uses crime records, community input, and predictive metrics to assess current area risk, based on severity, frequency, and crime volume.

Design a geolocation-based police dashboard to assess risk as a multi-dimensional object (severity, frequency, volume, required force) to support informed decision-making.

A community platform (partnered with Google/Apple Maps?) that fosters citizen engagement, keeping them informed about local activities and enabling verified crime reporting in their neighborhoods.

Create a resource optimization platform that suggests measures based on risk metrics and police experience with past crimes.

Evaluate outcomes based on citizen satisfaction and crime statistics.

Refine the prototype based on evaluation data and developing the platform in a demo stage.

Conclusion

The fuzzy-based crime prevention platform offers a dynamic, community-driven approach to urban safety. By using adaptive algorithms and real-time data, it enhances crime prediction and promotes fair, effective resource allocation, improving the trust between citizens, authorities and law enforcement.

Any Questions?

